PEOPLE COUNTER | PROJECT NOTE

Approach, observed limitation, and next steps

APPROACH

YOLOv8 detects people in each frame. Deep SORT links detections into tracks, while OSNet appearance embeddings help reconnect identities. A configured line and crossing confirmation logic generate IN/OUT events. The run saves an annotated video and an event CSV.

OBSERVED LIMITATION

The highlighted child was not counted as OUT: he disappeared for a long interval and was hidden behind a woman. The tracker did not maintain a reliable identity through that occlusion. Person IDs were also inconsistent across the clip, making event attribution less dependable.



Example frame supplied with the project; green box marks the child.

WHY IT HAPPENED

Long occlusions create gaps in detections and ambiguous re-identification. If a track is lost or assigned a new ID, the crossing logic may never receive a confirmed OUT for the original person.

WITH MORE TIME

Review the full clip and tune detection, tracking, and crossing thresholds. Test an occlusion-aware tracker and stronger long-term identity association. Label crossings and measure missed events, ID switches, and event precision/recall.

WHAT I LEARNED

Reliable people counting combines detection, multi-object tracking, identity association, camera geometry, and event logic. Re-identification helps, but does not guarantee identity continuity when people overlap or leave view.